5	**(a)**	An ideal gas is said to consist of molecules that are hard elastic identical spheres. State two further assumptions of the kinetic theory of gases.	
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		…………………………………………………………………………………………………	[2]
		Solution Any two of the following assumptions: Total volume of molecules negligible compared to the volume of vessel. No intermolecular forces between the molecules. Molecules are in random motion. Time of collision small compared with the time between collisions. Larger number of molecules.	 B1 B1
	(b)	The number of molecules per unit volume in an ideal gas is n. If it is assumed that all the molecules are moving with speed v_x in the x-direction, the pressure p exerted by the gas on the walls of the vessel is given by $$p = nm{v_x}^2$$ where m is the mass of one molecule. Explain the reasoning by which this expression is modified to give the formula $$p = \frac{1}{3}nm\langle c^2 \rangle.$$ where $\langle c^2 \rangle$ is the mean square speed of the molecules.	
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		…………………………………………………………………………………………………	[2]
		Solution In the actual scenario, there is <u>a range of velocities of molecules</u> in any given direction. Thus, the average of v^2 of the molecules should be considered. Molecules move in random direction and can move in x, y and z-direction. Since the <u>probability of the movement of molecules in either direction is the same</u>, $\langle {v_x}^2 \rangle = \langle {v_y}^2 \rangle = \langle {v_z}^2 \rangle$ $\langle {v_x}^2 \rangle = \frac{1}{3}\langle c^2 \rangle$	 B1 B1

	(c)	The density of an ideal gas is 1.2 kg m^{-3} at a pressure of 1.0×10^5 Pa and a temperature of 27 °C.		
		(i)	Calculate the root-mean-square (r.m.s.) speed of the molecules of the gas at 27 °C.	
			root-mean-square speed = ………………….……….. m s^{-1}	[3]
			Solution Nm = mass of gas $\rho = \frac{M}{V} = \frac{Nm}{V} = nm$ Therefore, $p = \frac{1}{3}nm\langle c^2\rangle = \frac{1}{3}\rho\langle c^2\rangle$ $1.0\times10^5 = \frac{1}{3}(1.2)\langle c^2\rangle$ $\langle c^2\rangle = 2.5\times10^5$ $C_{\text{r.m.s.}} = \sqrt{2.5\times10^5}$ = 500 m s^{-1}	C1 M1 A1
		(ii)	Calculate the mean-square speed of the molecules at 207 °C.	
			mean-square speed = ………………….………….. m^2 s^{-2}	[2]
			Solution $E = \frac{3}{2}kT = \frac{1}{2}m\langle c^2\rangle$ $\langle c^2\rangle \propto T$ $\frac{\langle c^2\rangle_{207^0 C}}{\langle c^2\rangle_{27^0 C}} = \frac{207+273.15}{27+273.15}$ $\langle c^2\rangle_{207^0 C} = \left(\frac{207+273.15}{27+273.15}\right)(2.5\times10^5)$ = 4.0×10^5 m^2 s^{-2}	M1 A1

[Total: 9]